

Phone to Humanoid: can a humanoid copy a squat from phone video?

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When a humanoid copies an athlete from ordinary phone video, which movements survive, and where do they break?



Bodyweight squat filmed on a phone at about 45°, 3 s in. Middle: a simulated Unitree G1 driven only by my joint angles has already fallen. Right: the same targets plus a simple ankle balance rule, still squatting. Faces and bystanders blurred.

Method

(1) MediaPipe Pose Landmarker (heavy) estimates 3D joints in every frame; the person was found in 100% of frames. (2) Knee, hip and trunk-lean angles are smoothed, split into reps and compared with dynamic time warping (the method from my final-year thesis). (3) The angles drive a Unitree G1 in MuJoCo (Menagerie model, 29 joints); I measured its joint sign conventions directly, solved the ankle so both feet stay flat, and kept the feet planted. (4) Physics replay: the G1's position actuators track those targets under gravity and contact.

Results

Stage	Result (45° clip, 8 full reps, 17 s)
Human consistency	Reps differ by about 2° after DTW alignment.
Pose replay, no physics	Every frame fits the G1's joint limits. Ankle needs up to 38.7° of shin-forward tilt (limit 50°); centre of mass stays over the feet, 2 cm margin at the bottom.
Physics, joint angles only	Falls forward at 1.2 s , bottom of the first squat (2.7 s with the robot's default arms).
Physics + ankle balance	Upright through all 8 squats (ankle target nudged by centre-of-mass drift; up to 17° of correction), but the feet shuffle back about 65 cm. Stricter contact settings do not change this.
Balance-aware retargeting	Tilting each pose so the centre of mass sits over mid-foot (about 2.5° extra lean) still falls without feedback.

What it means

The squat is kinematically feasible but dynamically unstable for the G1 when it simply copies joint angles. A one-line ankle strategy prevents the fall but not the foot shuffle; standing still and balanced needs a whole-body balance controller, the gap that current work on learning humanoid skills from human video closes with reinforcement learning. That is the problem I want to work on.

Limits

One athlete, one movement, one camera angle so far (a front view reads the squat about 14° deeper). Angles are zeroed at standing because one camera reads a straight knee as about 18° bent. Arms copy angles, not hand positions. The simulated G1 keeps Unitree's joint torque limits, but its gains are untuned and the balance rule reads the true centre of mass, which a real robot would have to estimate. Next: forward lunge, arm raise and instep kick with more athletes.

Video: [watch the three-panel clip \(Google Drive\)](#) · Code and notebook: github.com/samuelogundipe/phone2humanoid